Chapter 2. Topology of Euclidean Space

Open and closed sets, Bolzano–Weierstrass, Cauchy sequences, series, compactness, connectedness

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2.1 Euclidean space

Motivation. To study functions of several variables we need distance in $\mathbb{R}^n$. The Cauchy–Schwarz inequality is what makes the Euclidean distance satisfy the triangle inequality.

$\mathbb{R}^n$ is the set of n-tuples $x = (x_1, \dots, x_n)$ with coordinatewise addition and scalar multiplication, the **inner product** $\langle x, y\rangle = \sum x_i y_i$, the **norm** $\|x\| = \langle x, x\rangle^{1/2}$ and the **distance** $\|x - y\|$.

Theorem 2.1 (2.1.1, Cauchy–Schwarz inequality)

$|\langle x, y\rangle| \le \|x\|\|y\|$ for all $x, y \in \mathbb{R}^n$, with equality iff x, y are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $\|x - z\| \le \|x - y\| + \|y - z\|$.

Proof. If $y = 0$ both sides vanish. Otherwise, for real t, $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$; take $t = \langle x, y\rangle / \|y\|^2$ to get $0 \le \|x\|^2 - \langle x, y\rangle^2/\|y\|^2$. Equality forces $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Also $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, so a sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges.

2.2 Open and closed sets

Motivation. Many theorems need a domain without boundary points (to differentiate) or one containing all its limits (to guarantee maxima). Open and closed sets express these two requirements.

The **ϵ-neighbourhood** of x is $N(x, \epsilon) = \{y : \|y - x\| < \epsilon\}$. A point x is an **interior point** of A if some $N(x, \epsilon) \subset A$; A is **open** if all its points are interior. x is a **limit point** of A if every $N(x, \epsilon)$ contains a point of A other than x; A' denotes the set of limit points, and $\overline{A} = A \cup A'$ is the **closure**. A is **closed** if $\mathbb{R}^n - A$ is open. Points of $A - A'$ are **isolated points**. Every $N(x, \epsilon)$ is open, by the triangle inequality.

Proposition 2.2 (2.2.1)

(a) $\emptyset$ and $\mathbb{R}^n$ are open;
(b) unions of open sets are open;
(c) finite intersections of open sets are open. Dually, $\emptyset, \mathbb{R}^n$ are closed, intersections of closed sets are closed, and finite unions of closed sets are closed.

Proof. (a), (b) are immediate.

(c) If $x \in U_1 \cap \cdots \cap U_k$ with $N(x, \epsilon_i) \subset U_i$, then $N(x, \min \epsilon_i)$ lies in the intersection. The dual statements follow by De Morgan's laws. ∎

An infinite intersection of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.

Proposition 2.3 (2.2.2)

For $A \subset \mathbb{R}^n$ and $x \in \mathbb{R}^n$:
(a) $x \in A'$ iff some sequence in $A - \{x\}$ converges to x;
(b) $x \in \overline{A}$ iff some sequence in A converges to x.

Proof. (a) If $x \in A'$, pick $x_k \in A \cap N(x, 1/k)$, $x_k \ne x$. Conversely, if $x_k \to x$ with $x_k \in A - \{x\}$, every $N(x, \epsilon)$ contains some x_k.

(b) If $x \in A$ use the constant sequence; otherwise $x \in A'$ and use (a). Conversely, if $x_k \to x$, $x_k \in A$, either $x = x_k$ for some k or the x_k lie in $A - \{x\}$. ■

Theorem 2.4 (2.2.3)

For $F \subset \mathbb{R}^n$ the following are equivalent:
(a) F is closed;
(b) $F' \subset F$;
(c) $F = \overline{F}$.

Proof. (a)⇒(b): if $x \notin F$, some $N(x, \epsilon)$ misses F, so $x \notin F'$. (b)⇒(c): $\overline{F} = F \cup F' = F$. (c)⇒(a): if $x \notin F$, then $x \notin F'$, so some $N(x, \epsilon)$ contains no point of F (other than x, which is not in F); thus $\mathbb{R}^n - F$ is open. ■

Corollary 2.5 (2.2.4)

A' is closed for every $A \subset \mathbb{R}^n$; hence $\overline{A}$ is closed, and it is the smallest closed set containing A.

Proof. Let $x \in (A')'$ and $\epsilon > 0$. Choose $y \in A'$ with $0 < \|y - x\| < \epsilon$, and $\delta = \min(\|y - x\|, \epsilon - \|y - x\|) > 0$. Then $N(y, \delta)$ contains a point $z \in A$ with $z \neq y$; $z \in N(x, \epsilon)$, and $z \neq x$ because $\|z - y\| < \|x - y\|$. So $x \in A'$. The same argument shows $(\overline{A})' \subset A'$: if $N(x, \epsilon)$ contains $y \in A \cup A'$ with $y \neq x$, either $y \in A$, or $y \in A'$ and a small neighbourhood of y inside $N(x, \epsilon)$ contains a point of A other than x. So $\overline{A}$ is closed by Theorem 2.4. If $F \supset A$ is closed, $F \supset F' \supset A'$, so $F \supset \overline{A}$. ■

For $X \subset \mathbb{R}^n$, a set $U \subset X$ is **open in** X (relatively open) if for each $x \in U$ there is $\epsilon > 0$ with $N(x, \epsilon) \cap X \subset U$.

Proposition 2.6 (2.2.5)

For $U \subset X \subset \mathbb{R}^n$ the following are equivalent:
(a) U is open in X;
(b) $U = V \cap X$ for some open $V \subset \mathbb{R}^n$.

Proof. (a)⇒(b): for $x \in U$ choose ϵ_x with $N(x, \epsilon_x) \cap X \subset U$, and let $V = \bigcup_x N(x, \epsilon_x)$. (b)⇒(a): if $x \in V \cap X$ and $N(x, \epsilon) \subset V$, then $N(x, \epsilon) \cap X \subset U$. ■

Similarly, $F \subset X$ is closed in X iff $F = E \cap X$ with E closed in $\mathbb{R}^n$.

2.3 The Bolzano–Weierstrass theorem and Cauchy sequences

Motivation. Every bounded sequence has a convergent subsequence. This is the main way to produce limits out of nothing, and it shows that a sequence converges exactly when its terms eventually stay close to each other (the Cauchy criterion), which can be checked without knowing the limit.

A **box** is $B = I_1 \times \cdots \times I_n$ with bounded closed intervals I_k.

Lemma 2.7 (2.3.1, nested boxes)

If $B_1 \supset B_2 \supset \cdots$ are boxes, then $\bigcap_m B_m \neq \emptyset$.

Proof. Write $B_m = I_1^m \times \cdots \times I_n^m$. For each k, the intervals I_k^m are nested, so contain a common point x_k (Theorem 1.20); then $(x_1, \ldots, x_n) \in \bigcap B_m$. ■

Theorem 2.8 (2.3.2, Bolzano–Weierstrass)

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point.

Proof. A lies in a box B_1. Divide B_1 into 2^n boxes by bisecting each side; one of them, B_2, contains infinitely many points of A. Repeating, we obtain nested boxes B_m whose sides have length 2^{1-m} times those of B_1, each containing infinitely many points of A. Let $x \in \bigcap B_m$ (Lemma 2.7). Any $N(x, \epsilon)$ contains B_m for large m (its diameter tends to 0), hence infinitely many points of A; so $x \in A'$. ■

Proposition 2.9 (2.3.3)

If x is a limit point of A, then $A \cap N(x, \epsilon)$ is infinite for every $\epsilon > 0$.

Proof. If $A \cap N(x, \epsilon) - \{x\} = \{a_1, \ldots, a_k\}$, then $N(x, \delta)$ with $\delta = \min \|a_i - x\|$ contains no point of A other than x, a contradiction. ■

Theorem 2.10 (2.3.4)

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. If the range $\{x_k\}$ is finite, some value is taken infinitely often, giving a constant subsequence. Otherwise it is a bounded infinite set with a limit point x (Theorem 2.8). Each $N(x, 1/j)$ contains infinitely many terms (Proposition 2.9), so we can choose $k_1 < k_2 < \cdots$ with $\|x_{k_j} - x\| < 1/j$. ■

For a bounded real sequence (a_n) let A be the set of limits of its convergent subsequences.

Theorem 2.11 (2.3.5)

$\limsup a_n = \max A$ and $\liminf a_n = \min A$.

Proof. Let $a = \limsup a_n$. By Proposition 1.17(b) we may choose $n_1 < n_2 < \cdots$ with $a_{n_k} > a - 1/k$, and by (a) of the same proposition $a_{n_k} < a + 1/k$ for large k; so $a_{n_k} \to a$ and $a \in A$. If $a_{n_k} \to b$, then $a_{n_k} < a + \epsilon$ for large k, so $b \le a$. The $\liminf$ case is symmetric. ■

A sequence is **Cauchy** if for every $\epsilon > 0$ there is N with $\|x_m - x_n\| < \epsilon$ for all $m, n \ge N$.

Proposition 2.12 (2.3.6)

Every convergent sequence is Cauchy, and every Cauchy sequence is bounded.

Proof. If $x_n \to x$, choose N with $\|x_n - x\| < \epsilon/2$ for $n \ge N$; then $\|x_m - x_n\| < \epsilon$. If (x_n) is Cauchy, choose N for $\epsilon = 1$; then $\|x_n\| \le \max\{\|x_1\|, \ldots, \|x_N\|\} + 1$. ■

Lemma 2.13 (2.3.7)

If a subsequence of a Cauchy sequence (x_n) converges to x, then (x_n) converges to x.

Proof. Given ϵ, choose N with $\|x_m - x_n\| < \epsilon/2$ for $m, n \ge N$, and k with $n_k \ge N$ and $\|x_{n_k} - x\| < \epsilon/2$. For $n \ge N$, $\|x_n - x\| \le \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \epsilon$. ■

Theorem 2.14 (2.3.8, Cauchy criterion)

Every Cauchy sequence in $\mathbb{R}^n$ converges.

Proof. It is bounded (Proposition 2.12), so has a convergent subsequence (Theorem 2.10); apply Lemma 2.13. ■

Corollary 2.15 (2.3.9, Cauchy criterion for series)

$\sum a_n$ converges iff for every $\epsilon > 0$ there is N with $|a_{m+1} + \cdots + a_n| < \epsilon$ for all $n > m \geq N$.

Proof. Apply Theorem 2.14 to the partial sums s_n, since $s_n - s_m = a_{m+1} + \cdots + a_n$. ■

In particular, if $\sum a_n$ converges then $a_n \to 0$, and if $\sum |a_n|$ converges (**absolute convergence**) then $\sum a_n$ converges, because $|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n|$.

2.4 Series

Motivation. Infinite sums are limits of partial sums, so convergence of series is a special case of convergence of sequences. The tests below (comparison, ratio, root, alternating) are practical forms of the Cauchy criterion, and absolute convergence explains when terms may be rearranged.

Proposition 2.16 (2.4.1, rearrangement of nonnegative series)

Let $a_n \geq 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ a bijection. If $\sum a_n = s$, then $\sum a_{\tau(n)} = s$; if $\sum a_n$ diverges, so does $\sum a_{\tau(n)}$.

Proof. Each partial sum $\sum_{n \leq N} a_{\tau(n)}$ is at most $\sum_{n \leq M} a_n \leq s$ with $M = \max \tau(\{1, \ldots, N\})$, so $t = \sum a_{\tau(n)} \leq s$ by Corollary 1.16. Applying this to τ^{-1} gives $s \leq t$. The divergence statement follows by symmetry. ■

More generally, an absolutely convergent series may be rearranged without changing its sum: split a_n into positive and negative parts $a_n^{\pm} = \max(\pm a_n, 0)$, both summable, and apply the proposition to each. By contrast, a conditionally convergent series can be rearranged to have any sum (Riemann).

Theorem 2.17 (2.4.2, alternating series test)

If $x_1 \geq x_2 \geq \cdots \geq 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and its sum s satisfies $|s - s_n| \leq x_{n+1}$.

Proof. $s_{2m} = (x_1 - x_2) + \cdots + (x_{2m-1} - x_{2m})$ increases, $s_{2m+1} = x_1 - (x_2 - x_3) - \ldots$ decreases, and $s_{2m} \leq s_{2m+1}$. Both converge (Theorem 1.15), and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$, so they have the same limit s, which lies between consecutive partial sums. ■

Theorem 2.18 (2.4.3, comparison test)

If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.

Proof. The partial sums of $\sum a_n$ are bounded by $\sum b_n$; use Corollary 1.16. ■

Proposition 2.19 (2.4.4, root test)

Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, $\sum a_n$ diverges.

Proof. If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large n (Proposition 1.17), and $\sum \beta^n < \infty$. If $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$. ■

Proposition 2.20 (2.4.5, ratio test)

Let $a_n \neq 0$. If $\limsup |a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely; if $|a_{n+1}/a_n| \geq 1$ for all large n, then $\sum a_n$ diverges.

Proof. If $|a_{n+1}/a_n| \leq \beta < 1$ for $n \geq N$, then $|a_n| \leq |a_N|\beta^{n-N}$. In the second case $|a_n|$ is eventually nondecreasing and positive, so $a_n \not\to 0$. ■

Proposition 2.21 (2.4.6)

If $c_n > 0$, then $\liminf \frac{c_{n+1}}{c_n} \leq \liminf c_n^{1/n} \leq \limsup c_n^{1/n} \leq \limsup \frac{c_{n+1}}{c_n}$.

Proof. Let $\beta = \limsup c_{n+1}/c_n$, assumed finite, and $\gamma > \beta$. For $n \geq N$, $c_{n+1} < \gamma c_n$, so $c_n < c_N \gamma^{-N} \gamma^n$ and $c_n^{1/n} < (c_N \gamma^{-N})^{1/n} \gamma \to \gamma$. Hence $\limsup c_n^{1/n} \leq \gamma$ for all $\gamma > \beta$. The left inequality is analogous. ■

So the root test is stronger than the ratio test: whenever the ratio test decides, so does the root test.

2.5 Compact sets

Motivation. A closed bounded set in $\mathbb{R}^n$ has the property that every open cover has a finite subcover. This abstract property is what lets us pass from local estimates to global ones, and Heine–Borel shows it is the same as being closed and bounded.

An **open cover** of K is a family of open sets whose union contains K; K is **compact** if every open cover has a finite subcover.

Proposition 2.22 (2.5.1)

Every compact set is bounded and closed.

Proof. The sets $N(0,k)$ cover K; finitely many do, so $K \subset N(0,m)$. If $x \notin K$, the open sets $U_k = \{y : \|y - x\| > 1/k\}$ cover K; a finite subcover gives $K \subset U_m$, so $N(x, 1/m)$ misses K. Hence $\mathbb{R}^n - K$ is open. ■

Theorem 2.23 (2.5.2)

Every box $B = I_1 \times \cdots \times I_n$ is compact.

Proof. Suppose an open cover $\{U_i\}$ of B has no finite subcover. Bisecting all sides, one of the 2^n sub-boxes, B_2, is not covered by finitely many U_i; continuing gives nested boxes B_m of diameter $\operatorname{diam} B_1/2^{m-1}$ none finitely covered. Let $x \in \bigcap B_m$ (Lemma 2.7) and $x \in U_i$, $N(x,\epsilon) \subset U_i$. For large m, $B_m \subset N(x,\epsilon) \subset U_i$, a contradiction. ■

Proposition 2.24 (2.5.3)

A closed subset F of a compact set K is compact.

Proof. If $\{U_i\}$ covers F, then $\{U_i\} \cup \{\mathbb{R}^n - F\}$ covers K; a finite subcover, with $\mathbb{R}^n - F$ removed, covers F. ■

Theorem 2.25 (2.5.4, Heine–Borel)

For $K \subset \mathbb{R}^n$ the following are equivalent:
(a) K is compact;
(b) K is bounded and closed;
(c) every infinite subset of K has a limit point in K.

Proof. (a)⇒(b): Proposition 2.22. (b)⇒(a): K lies in a box, which is compact (Theorem 2.23); apply Proposition 2.24. (b)⇒(c): Bolzano–Weierstrass gives a limit point, which is in K as K is closed. (c)⇒(b): if K were unbounded, choose $x_k \in K$ with $\|x_k\| > k$; the infinite set $\{x_k\}$ has no limit point at all. If $x \in K'$, Proposition 2.9 gives distinct $x_k \in K$ with $x_k \to x$; the set $\{x_k\}$ has x as its only limit point, so $x \in K$ by (c), and K is closed. ■

So K is compact iff every sequence in K has a subsequence converging in K.

Theorem 2.26 (2.5.5, finite intersection property)

If $\{K_i : i \in I\}$ are compact sets such that $\bigcap_{i\in J} K_i \neq \emptyset$ for every finite $J \subset I$, then $\bigcap_{i\in I} K_i \neq \emptyset$.

Proof. Fix i_0. If the intersection were empty, the open sets $\mathbb{R}^n - K_i$ would cover K_{i_0}, and finitely many $\mathbb{R}^n - K_{i_1}, \dots, \mathbb{R}^n - K_{i_m}$ would cover it; then $K_{i_0} \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

In particular a decreasing sequence of nonempty compact sets has nonempty intersection.

Corollary 2.27 (2.5.6)

A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point.

Proof. Let $S = \{s_1, s_2, \dots\}$ and suppose no point is isolated. Choose a closed ball $K_1 = \overline{N(y_1, r_1)}$ with $y_1 \in S$; inductively, since y_k is not isolated, $N(y_k, r_k)$ contains infinitely many points of S, so we can choose $y_{k+1} \in S \cap N(y_k, r_k)$, $y_{k+1} \neq s_k$, and a closed ball K_{k+1} about y_{k+1} inside $N(y_k, r_k)$ not containing s_k. The sets $K_k \cap S$ are nonempty, compact and decreasing, so they have a common point (Theorem 2.26); it lies in S but differs from every s_k, a contradiction. ■

Thus a nonempty closed set without isolated points (a **perfect set**), such as the Cantor set, is uncountable.

2.6 Connected sets

Motivation. Intervals are the sets that cannot be split into two separated pieces. This property is what lies behind the intermediate value theorem.

$S \subset \mathbb{R}^n$ is **disconnected** if there are open U, V with

$$U \cap S \neq \emptyset, \quad V \cap S \neq \emptyset, \quad U \cap V \cap S = \emptyset, \quad S \subset U \cup V,$$

and **connected** otherwise. A set $S \subset \mathbb{R}$ is an **interval** if $x, y \in S$ and $x < z < y$ imply $z \in S$.

Proposition 2.28 (2.6.1)

A subset S of $\mathbb{R}$ is connected iff it is an interval.

Proof. If S is not an interval, take $x < z < y$ with $x, y \in S$, $z \notin S$; then $U = (-\infty, z)$, $V = (z, \infty)$ disconnect S. Conversely let S be an interval and U, V as above, with $x \in U \cap S$, $y \in V \cap S$, say $x < y$; then $[x, y] \subset S$. Let $c = \sup([x, y] \cap U)$. If $c \in U$, then $c < y$ (as $y \notin U$) and U contains $[c, c+\delta)$, contradicting the definition of c; if $c \in V$, then $c > x$ and $(c-\delta, c] \subset V$, so $(c-\delta, c]$ misses U and c is not the supremum. But $c \in [x, y] \subset S \subset U \cup V$, a contradiction. ■

Proposition 2.29 (2.6.2)

If $\{C_i\}$ are connected and $\bigcap C_i \neq \emptyset$, then $C = \bigcup C_i$ is connected.

Proof. Suppose U, V disconnect C, and let $p \in \bigcap C_i$, say $p \in U$. Each C_i is connected and meets U (at p), so $C_i \cap V = \emptyset$ (otherwise U, V would disconnect C_i). Then $C \cap V = \emptyset$, a contradiction. ■

Convex sets, in particular balls and boxes, are connected: they are unions of segments through a common point, each segment being connected as a continuous image of an interval (Theorem 3.13).

2.7 Selected exercises

Motivation. Three problems from the book's exercise set: the diameter gives a convenient reformulation of the Cauchy condition and of the nested set theorems, and two short arguments show how rigid connectedness is.

For a nonempty $X \subset \mathbb{R}^n$ let $\operatorname{diam} X = \sup\{\|x - y\| : x, y \in X\} \in [0, \infty]$.

Exercise 2.7.23 (Diameter)

(a) Show that $\operatorname{diam} \overline{X} = \operatorname{diam} X$ for every nonempty $X \subset \mathbb{R}^n$.
(b) Show that a sequence (x_k) in $\mathbb{R}^n$ is Cauchy iff $\lim_{N\to\infty} \operatorname{diam}\{x_k : k \geq N\} = 0$.
(c) Let $X_1 \supset X_2 \supset \cdots$ be nonempty closed bounded sets with $\operatorname{diam} X_k \to 0$. Show that $\bigcap_k X_k$ consists of exactly one point.

Solution.

(a)

Step 1. $X \subset \overline{X}$ gives $\operatorname{diam} X \leq \operatorname{diam} \overline{X}$. We may assume $\operatorname{diam} X < \infty$.

Step 2. Let $x', y' \in \overline{X}$ and $\epsilon > 0$. By Proposition 2.3(b) there are $x, y \in X$ with $\|x - x'\| < \epsilon$ and $\|y - y'\| < \epsilon$. Then

$$\|x' - y'\| \leq \|x' - x\| + \|x - y\| + \|y - y'\| < \operatorname{diam} X + 2\epsilon.$$

As ϵ is arbitrary, $\|x' - y'\| \leq \operatorname{diam} X$, and taking the supremum, $\operatorname{diam} \overline{X} \leq \operatorname{diam} X$.

(b) Let $d_N = \operatorname{diam}\{x_k : k \geq N\}$; it is decreasing in N, since the sets shrink.

Step 1. If (x_k) is Cauchy and $\epsilon > 0$, choose N with $\|x_m - x_k\| < \epsilon/2$ for $m, k \geq N$. Then $d_M \leq d_N \leq \epsilon/2 < \epsilon$ for all $M \geq N$, so $d_N \to 0$.

Step 2. Conversely, if $d_N \to 0$ and $\epsilon > 0$, choose N with $d_N < \epsilon$; for $m, k \geq N$, $\|x_m - x_k\| \leq d_N < \epsilon$.

(c)

Step 1. Each X_k is compact by the Heine–Borel theorem (Theorem 2.25), and the X_k are nonempty and decreasing, so $\bigcap_k X_k \neq \emptyset$ by Theorem 2.26.

Step 2. If $x, y \in \bigcap_k X_k$, then $\|x-y\| \le \operatorname{diam} X_k$ for every k, so $\|x-y\| = 0$ and $x = y$.

The hypothesis $\operatorname{diam} X_k \to 0$ is needed for uniqueness ($X_k = [0,1]$), and closedness for existence ($X_k = (0, 1/k)$). ■

Exercise 2.7.25 (Sets between a connected set and its closure)

Let $A \subset \mathbb{R}^n$ be connected and $A \subset B \subset \overline{A}$. Show that B is connected. In particular $\overline{A}$ is connected.

Solution.

Step 1. Suppose open U, V disconnect B: $U \cap B \neq \emptyset$, $V \cap B \neq \emptyset$, $U \cap V \cap B = \emptyset$, $B \subset U \cup V$.

Step 2. Since $A \subset B$, we get $U \cap V \cap A = \emptyset$ and $A \subset U \cup V$. As A is connected, U and V cannot both meet A; say $U \cap A = \emptyset$.

Step 3. Pick $x \in U \cap B$. Since U is open, $N(x, \epsilon) \subset U$ for some $\epsilon > 0$. Since $x \in B \subset \overline{A}$, Proposition 2.3(b) gives a point of A in $N(x, \epsilon) \subset U$, contradicting $U \cap A = \emptyset$. Hence B is connected. ■

Exercise 2.7.26 (Connected sets have no isolated points)

Let $A \subset \mathbb{R}^n$ be connected with at least two points. Show that every point of A is a limit point of A.

Solution.

Step 1. Suppose $x \in A$ is not a limit point of A: there is $\epsilon > 0$ with $N(x, 2\epsilon) \cap A = \{x\}$. Let

$$U = N(x, \epsilon), \qquad V = \{y \in \mathbb{R}^n : \|y - x\| > \epsilon\}.$$

U is open, and V is open because $y \mapsto \|y - x\|$ is continuous (or directly: if $\|y - x\| > \epsilon$, then $N(y, \|y - x\| - \epsilon) \subset V$ by the triangle inequality). Also $U \cap V = \emptyset$.

Step 2. $x \in U \cap A$. Let $z \in A$, $z \neq x$ (it exists by assumption). Then $z \notin N(x, 2\epsilon)$, so $\|z - x\| \ge 2\epsilon > \epsilon$ and $z \in V \cap A$.

Step 3. Every $z \in A$ lies in $U \cup V$: either $z = x \in U$, or $\|z - x\| \ge 2\epsilon$ and $z \in V$. So U, V disconnect A, a contradiction. ■